Render the Molecule: manuscript draft

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Title page

Title: Render the Molecule: zero-installation, path-traced images of Protein Data Bank entries with Blender and Molecular Nodes

Running title: Path-traced PDB images from an ID

Article type: Software / tools article (fits Protein Science "Tools for Protein Science"; condensable to a Bioinformatics Application Note of about 2,600 words)

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Candidate journal	Fit	Check before submitting
Protein Science (Tools for Protein Science)	Publishes Mol*, ChimeraX and RCSB tool papers	Word limit and figure count
Bioinformatics (Application Note)	Published NGL, 3Dmol.js, iCn3D and The Protein Imager	2,600 words, one figure
Journal of Molecular Biology (Computation Resources)	Published EzMol	Scope statement
Nucleic Acids Research (Web Server issue)	Published Mol* and NGL	The mandatory name and email field may count as a login requirement

Abstract

Motivation: Path-traced images of macromolecules convey shape and depth through physically based lighting, but making them has meant installing and learning professional 3D software.

Results: We present Render the Molecule, a web application, command-line tool and Google Colab notebook that renders any Protein Data Bank (PDB) entry from its identifier with Blender's Cycles path tracer through Molecular Nodes. It offers six representations, four colour schemes and biological assemblies, and embeds the software versions, source entry and references in every PNG file. On ten entries spanning 327 to 2.44 million atoms, every image of an entry up to 58,870 atoms, a size range that covers 97.9% of the PDB, rendered in 16–58 s on four CPU threads with at most 3.6 GB of memory.

The default preset reached a structural similarity of at least 0.996 against 512-sample references, and repeated renders were byte-identical across processes and thread counts, which makes content-addressed caching exact. Memory, not time, limited the largest entries: a ribosome surface needed 10.2 GB and a 2.44-million-atom capsid exhausted 15 GB.

Availability and implementation: Free and open source under GPL-3.0-or-later at github.com/vishalbhoir18/renderthemolecule, archived at doi:10.5281/zenodo.23960655. Web application: vishalbhoir18--render-the-molecule.us-east.modal.run.

Supplementary information: Benchmark scripts, per-render results and all benchmark images are in the repository's paper folder.

1. Introduction

Images of three-dimensional macromolecular structures are how most readers first meet a structural result. They appear in research figures, graphical abstracts, textbooks, grant proposals and outreach material, and they shape how a structure is understood well beyond the community that solved it (O'Donoghue et al., 2010; Johnson & Hertig, 2014; Goodsell & Jenkinson, 2018; Martinez et al., 2019; Torrez et al., 2026). The supply of structures to depict keeps growing. On 5 November 2026 the RCSB Protein Data Bank (PDB) listed 262,922 released entries, and RCSB.org now serves more than one million computed structure models next to them (Berman et al., 2000; Bittrich et al., 2023; Varadi et al., 2024).

The software used to make these images falls into two broad groups. Desktop programs such as VMD, PyMOL and UCSF ChimeraX combine analysis with high-quality rendering, including ray tracing and ambient occlusion (Humphrey et al., 1996; Goddard et al., 2018; Pettersen et al., 2021). Browser viewers built on WebGL, among them JSmol, NGL, 3Dmol.js, LiteMol, iCn3D and Mol*, removed installation altogether and made interactive viewing routine on structure databases (Hanson et al., 2013; Rose & Hildebrand, 2015; Rego & Koes, 2015; Sehnal et al., 2017; Rose et al., 2018; Wang et al., 2020; Sehnal et al., 2021). Real-time engines approximate global illumination in screen space to stay interactive (Rose et al., 2026). Lighting is not decoration: ambient occlusion and related shadowing cues measurably improve the perception of molecular shape, pockets and depth (Tarini et al., 2006; Wahle & Wriggers, 2015; Kozlíková et al., 2017).

Physically based path tracing, which solves the rendering equation by Monte Carlo sampling of light paths (Kajiya, 1986), gives these cues without approximation. It has been available to structural biologists for decades through programs such as Raster3D (Merritt & Bacon, 1997), and in illustrative form through Goodsell's non-photorealistic style (Goodsell et al., 2019). General-purpose 3D suites, Blender in particular, offer production path tracers, but setting up a molecular scene in them is laborious even for experts (Durrant, 2022). A series of bridges has lowered that barrier: ePMV embedded molecular modelling in animation packages (Johnson et al., 2011), BioBlender added surface properties and motion (Andrei et al., 2012), and BlendMol and Pyrite imported VMD, PyMOL and molecular dynamics scenes into Blender (Rajendiran & Durrant, 2018; Durrant, 2019). Molecular Nodes now builds molecular geometry natively inside Blender from PDB, mmCIF and trajectory files using Blender's procedural Geometry Nodes system (Johnston & Molecular Nodes contributors, 2026). Each of these tools still requires installing Blender and learning its interface.

A few services render on a server instead. EzMol guides users through a short wizard to an image (Reynolds et al., 2018), The Protein Imager couples an online viewer to automated high-quality server-side rendering (Tomasello et al., 2020), and Prot2Prot imitates photorealistic styles with a neural network that runs in the browser (Durrant, 2022). To our knowledge, no service yet returns a path-traced Blender image from nothing more than a PDB identifier, and none reports what such images cost to compute across the size range of the PDB.

Images also travel without their provenance. A figure rarely records which program, version and settings produced it, although software citation principles and the FAIR principles for research software ask for exactly that (Smith et al., 2016; Katz et al., 2021; Barker et al., 2022).

Here we describe Render the Molecule, a free, open-source web application, command-line tool and Google Colab notebook that turns a PDB identifier into a path-traced image rendered by Blender and Molecular Nodes. Every image carries its credits and a ready-made citation in its file metadata. We benchmark the tool on ten PDB entries spanning 327 to 2.44 million deposited atoms and report render time, peak memory, image quality against a high-sample reference, repeatability and the limits of the current deployment.

2. Materials and methods

2.1 Design goals

Render the Molecule was built around five requirements: (i) no installation on the user's side; (ii) publication-quality output from default settings; (iii) a complete and machine-readable record of provenance in every image; (iv) bit-for-bit reproducible renders for a given software version and option set; and (v) an operating cost low enough for a free public service.

2.2 Architecture and request flow

The web application is written in Python 3.13 with FastAPI (Figure 1). A request names a PDB identifier and up to ten options. Pydantic models validate it against fixed option lists and accept both classic four-character identifiers and the extended wwPDB format (pdb_0000xxxx). The validated options, together with a render-version string, are serialised to canonical JSON and hashed with SHA-256. The digest names the output file, so identical requests map to one image: a cached image is returned at once, and a request identical to one already queued joins that job instead of creating a new one.

Uncached jobs enter an in-memory first-in, first-out queue served by a single worker thread. One worker is deliberate, because the path tracer already uses every available CPU core. The queue holds at most 20 jobs and each client at most three active jobs. The browser polls a status endpoint and shows the user's position in the queue. A command-line interface and a Google Colab notebook call the same rendering code; the notebook runs on Google's free cloud machines, a route that ColabFold showed can bring demanding structural software to users without local hardware (Mirdita et al., 2022).

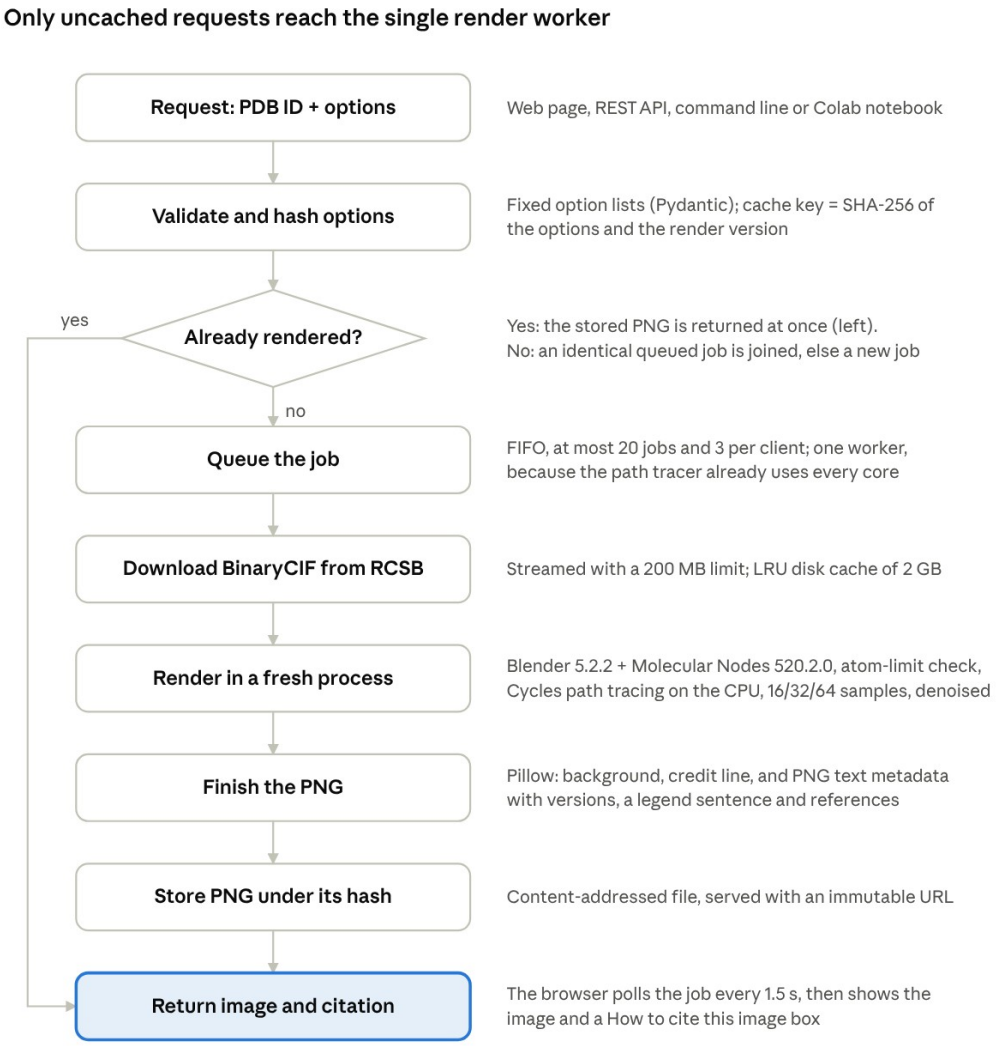


Figure 1 · request flow of Render the Molecule v0.1.0, from app/main.py, app/jobs.py and app/renderer.py

Figure 1. Request flow. Options are validated and hashed; a cached image is returned at once (left branch), and only new requests are queued, downloaded and path-traced in a fresh Blender process.

2.3 Structure retrieval

Coordinates are downloaded from the RCSB model server in BinaryCIF, a compact binary encoding of PDBx/mmCIF (Sehnal et al., 2020; Westbrook et al., 2022). Downloads are streamed with a 200 MB limit, decompressed if gzip-encoded, and written atomically. Structures and renders are kept in two disk caches of 2 GB each, pruned least-recently-used first.

2.4 Scene construction and rendering

Each job runs in a new Python process that imports Blender 5.2.2 (Blender Online Community, 2026) as the bpy module and Molecular Nodes 520.2.0 (Johnston & Molecular Nodes contributors, 2026). No Blender executable or add-on installation is involved. Running every render in its own process isolates the web server from Blender's global state, memory growth and crashes. Molecular Nodes parses the file with Biotite (Kunzmann & Hamacher, 2018) and MDAnalysis (Michaud-Agrawal et al., 2011) and builds geometry with Geometry Nodes.

Users choose one of six representations (cartoon, ribbon, molecular surface, spheres, ball-and-stick, sticks), four colour schemes (by chain, rainbow, secondary structure, and a Goodsell-like scheme with soft per-chain colours and darkened carbons), four materials, seven viewpoints, four output sizes, three quality levels and three backgrounds. Water is hidden in all-atom representations. For cartoon and ribbon, non-polymer, non-water atoms (ligands, cofactors and ions) are overlaid as ball-and-stick. Biological assemblies are built from the first assembly's symmetry operators in the mmCIF file (Krissinel & Henrick, 2007).

Before any geometry is built, the atom count after assembly expansion is checked against a per-representation limit: 3,000,000 atoms for cartoon and ribbon, 1,000,000 for surface and spheres, and 300,000 for ball-and-stick and sticks. Requests above the limit are refused with a message that suggests a cheaper option.

Images are rendered with Blender's Cycles path tracer on the CPU, with the settings Molecular Nodes applies: adaptive sampling with a noise threshold of 0.01, up to 12 light bounces (4 diffuse, 4 glossy), Intel Open Image Denoise guided by albedo and normal passes, the AgX view transform and a fixed random seed. The three quality levels cap sampling at 16, 32 and 64 samples per pixel. The film is transparent, and the chosen background is composited afterwards, so lighting is identical whatever the background colour. The number of render threads is read from the container's cgroup CPU quota rather than the host's CPU count.

2.5 Provenance and citation

Each image carries credits in two forms. A small credit line (for example, "PDB 4HHB · Molecular Nodes · Render the Molecule by Vishal Bhoir") is drawn in the lower right corner, scaled to the image width; users may switch it off. PNG text chunks always record the title, the software versions, the source entry with its RCSB link, a one-sentence legend for a figure caption, and APA-formatted references to the tool, Molecular Nodes, Blender, the PDB and the primary publication of the entry. The same text is shown after each render in a "How to cite this image" box. The software is archived on Zenodo (Bhoir, 2026) and described by a CITATION.cff file, following software citation principles (Smith et al., 2016; Katz et al., 2021).

2.6 Deployment and testing

The application ships as a Docker image based on python:3.13-slim with Mesa libraries, so Blender starts without a GPU (Boettiger, 2015). The public instance runs on Modal as a single container with two physical CPU cores and 8 GB of memory, which starts on the first request and stops 300 s after the last. All Python dependencies are pinned to exact versions. The test suite covers validation, caching, the job

queue, the citation text, the notebook and a real render of PDB 1CRN, and continuous integration builds and starts the container on every push. Users give a name and email address before rendering. The operator may record these with each render in a private Google Sheet; they are never part of the image cache key or of shared links.

2.7 Benchmark protocol

All measurements were made on one Linux virtual machine (Intel Xeon processor at 2.80 GHz, 4 vCPUs, 15 GB RAM, Linux 6.18, Python 3.13.18) with the pinned software of release v0.1.0. A benchmark script (paper/benchmark/run_benchmark.py in the repository) downloads each structure with the application's own code and runs the application's own render module in a fresh process for each image, exactly as the web server does. Unless varied, every render used the defaults: 1280 × 720 pixels, Standard quality (32 samples), ambient-occlusion material, default viewpoint, colour by chain, white background, ligands shown and credit line on, with 4 render threads.

The panel comprised ten PDB entries chosen to span four orders of magnitude in size, three experimental methods and protein, DNA and RNA (Table 1).

Table 1. Benchmark panel. Deposited atom counts, methods and resolutions from the RCSB Data API on 5 November 2026.

PDB ID	Molecule	Method	Resolution (Å)	Deposited atoms
1CRN	Crambin	X-ray	1.50	327
1BNA	B-DNA dodecamer	X-ray	1.90	566
1UBQ	Ubiquitin	X-ray	1.80	660
1EHZ	Yeast tRNA-Phe	X-ray	1.93	1,821
4HHB	Human deoxyhaemoglobin	X-ray	1.74	4,779
1CD3	Bacteriophage φX174 procapsid (asymmetric unit)	X-ray	3.50	9,851
6VXX	SARS-CoV-2 spike, closed state	Cryo-EM	2.80	23,694
1AON	GroEL–GroES–ADP chaperonin	X-ray	3.00	58,870
4UG0	Human 80S ribosome	Cryo-EM	3.60	218,776
3J3Q	Complete HIV-1 capsid	Cryo-EM + MD model	n/a	2,440,800

Seven experiments were run. E1 rendered every entry as cartoon, surface and spheres to measure how cost scales with size. E2 rendered haemoglobin (4HHB) in all six representations and in three colour schemes. E3 rendered 4HHB (cartoon) and 1AON (surface) at the three quality levels and at 512 samples as a reference. E4 varied the number of render threads (1, 2 and 4). E5 rendered the φX174 procapsid

with and without its 60-copy biological assembly. E6 rendered the same request twice to test repeatability, and E7 varied the output size.

For each render we recorded the scene set-up time and the path-tracing time reported by the renderer, the wall-clock time of the whole process (including interpreter start and import of Blender), and the peak resident memory of the process from the operating system (wait4), reported in GB of 1,024 MB. Image quality in E3 was measured as the structural similarity index (SSIM; Wang et al., 2004) and peak signal-to-noise ratio (PSNR) of the luminance channel against the 512-sample reference, excluding the bottom 6% of rows that hold the identical credit line. Repeatability in E6 was assessed by pixel-wise comparison of the two PNG files.

3. Results

3.1 One request gives a finished figure

From its identifier alone, the default request rendered human deoxyhaemoglobin (4HHB) in 33.7 s of wall-clock time on four CPU threads. The image showed the four chains in separate colours, with the haem groups and other ligands overlaid in ball-and-stick (Figure 2A). All six representations and four colour schemes of the same entry completed in 22–40 s of path tracing each, with peak memory between 1.85 and 2.23 GB (Table 2; Figure 2). Spheres were the cheapest representation and ball-and-stick the most expensive. The colour scheme changed path-tracing time by less than 1 s for the cartoon (27.9–28.8 s) and by at most 2.7 s for the surface (31.6–34.3 s).

Every wall-clock time contained a fixed overhead: about 3 s to start the interpreter and import Blender, and 2.0–3.5 s to build the scene. For a typical entry, therefore, more than 80% of the time is spent in the path tracer itself.

Table 2. Cost of each representation for haemoglobin (4HHB, 4,779 atoms) at 1280 × 720, Standard quality, 4 threads, colour by chain (experiment E2).

Representation	Path tracing (s)	Wall clock (s)	Peak memory (GB)	PNG size (kB)
Cartoon	28.1	33.7	1.98	459
Ribbon	30.1	35.8	1.97	474
Surface	31.6	37.7	2.00	377
Spheres	22.1	27.2	1.85	621
Ball and stick	39.7	45.5	2.10	664
Sticks	28.9	34.0	2.23	596

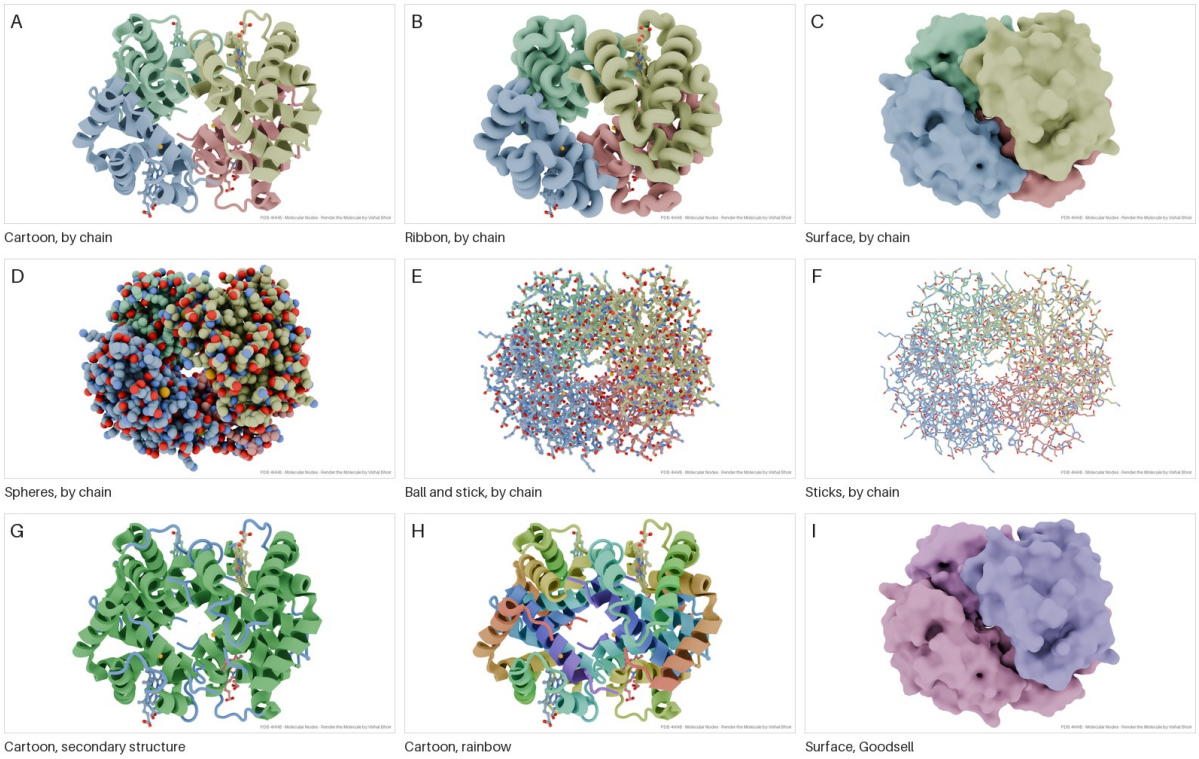


Figure 2. Haemoglobin (PDB 4HHB) in every representation and colour scheme, each panel one unedited output (1280 × 720, Standard quality, default viewpoint) with its credit line. (A–F) Cartoon, ribbon, surface, spheres, ball-and-stick and sticks, coloured by chain. (G) Cartoon coloured by secondary structure. (H) Cartoon, rainbow. (I) Surface, Goodsell-like scheme.

3.2 Cost grows slowly with size until memory runs out

Wall-clock time per image grew far more slowly than structure size (Figure 3A; experiment E1). From crambin (327 atoms) to the human 80S ribosome (218,776 atoms), a 669-fold increase, the cartoon took 17.8 s and 57.3 s, and spheres 19.6 s and 42.1 s. Every image of every entry up to the GroEL–GroES complex (58,870 atoms) finished within one minute in all three representations. Time tracked how much of the frame the molecule filled more closely than its atom count: haemoglobin, which fills the frame, took longer as a cartoon (35.3 s) than the much larger SARS-CoV-2 spike (28.0 s).

Memory, not time, set the limits (Figure 3B). Cartoon and sphere images peaked at no more than 3.0 GB for every entry up to the ribosome. Surfaces grew fastest: 1.9 GB for crambin, 3.6 GB for GroEL–GroES, and 10.2 GB for the ribosome, whose surface took 175 s. The ribosome surface therefore exceeds the 8 GB of the public deployment, although it is far below the 1,000,000-atom limit for surfaces.

The complete HIV-1 capsid (3J3Q, 2.44 million atoms) exposed both limits. Its cartoon, which the 3,000,000-atom limit allows, was stopped by the operating system's out-of-memory killer after 278 s, at 13.2 GB on the 15 GB benchmark machine. Its surface and spheres were correctly refused by the atom limit, but only after 247 s of parsing at 4.0 GB, because the atom count is checked after the structure has been loaded.

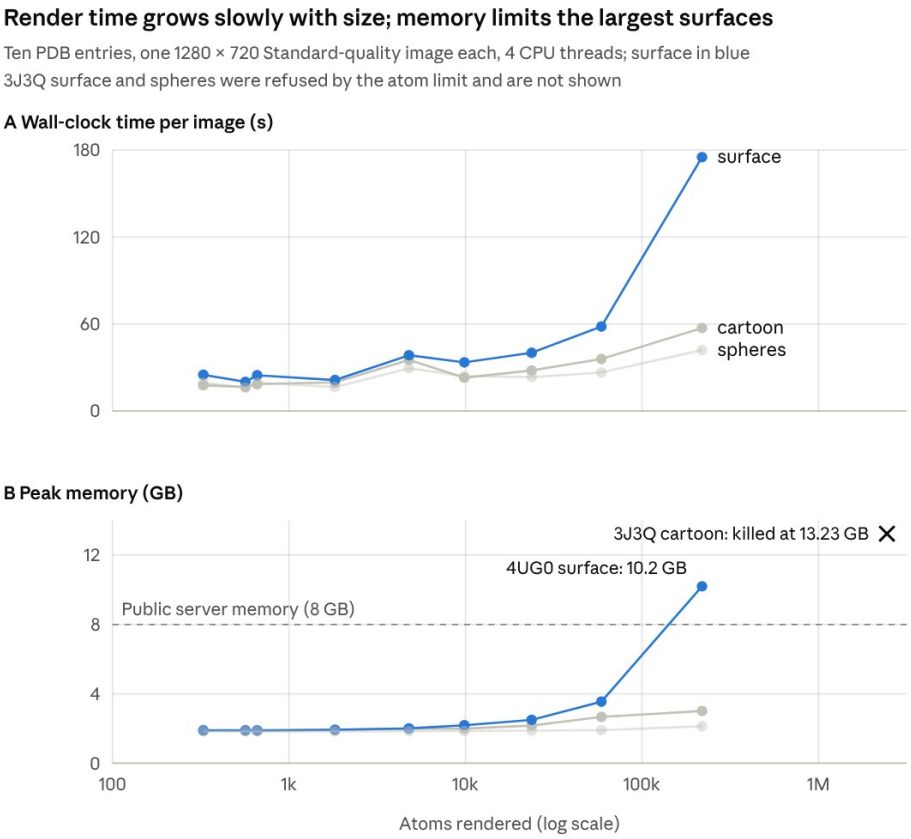


Figure 3 · experiment E1, paper/benchmark/results.csv · 4 vCPUs, 15 GB RAM; atoms for 3J3Q are its deposited count

Figure 3. Cost of one image against structure size. (A) Wall-clock time per image, including process start and scene set-up. (B) Peak resident memory of the render process; the dashed line marks the memory of the public deployment. The cross marks the 3J3Q cartoon, killed for lack of memory.

Representative images from the panel are shown in Figure 4, each one unedited output of a single request.

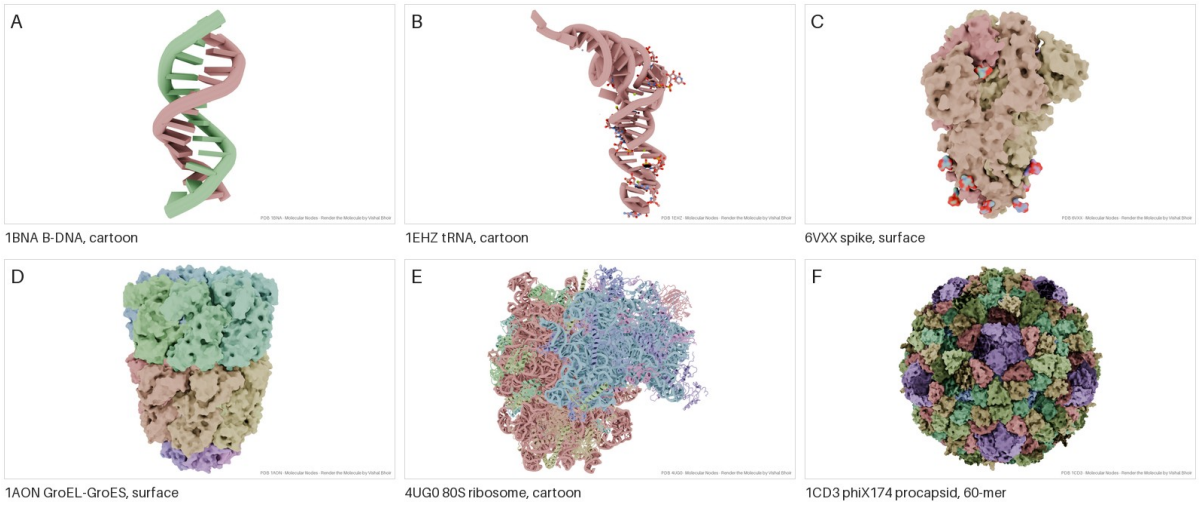


Figure 4. Benchmark images, unedited: (A) B-DNA dodecamer (1BNA), cartoon; (B) yeast tRNA-Phe (1EHZ), cartoon; (C) SARS-CoV-2 spike in the closed state (6VXX), surface; (D) GroEL–GroES chaperonin (1AON), surface; (E) human 80S ribosome (4UG0), cartoon; (F) φX174 procapsid (1CD3) as its 60-copy biological assembly, surface.

3.3 Biological assemblies cost time, not memory

Expanding the bacteriophage φX174 procapsid (1CD3) from its asymmetric unit to the 60-copy biological assembly multiplied the atom count 60-fold, from 9,851 to 591,060 (experiment E5). Path-tracing time rose only 3.6-fold for the cartoon (17.8 to 64.5 s) and 2.8-fold for the surface (26.1 to 72.3 s). Peak memory did not rise at all: 2.00 versus 2.02 GB for the cartoon, and 2.14 versus 2.08 GB for the surface. This is consistent with the assembly being drawn as transformed instances of one set of geometry, so memory follows the asymmetric unit while time follows the area and complexity of what the camera sees.

3.4 Few samples suffice once the image is denoised

Cycles stops sampling a pixel once its estimated noise falls below a threshold, and the result is then denoised. The quality presets therefore set an upper bound on samples per pixel, not a fixed cost. Against 512-sample references, the Standard preset (32 samples) reached a structural similarity over the molecule's pixels of 0.996 for the haemoglobin cartoon and 0.997 for the GroEL–GroES surface (Figure 5). Each doubling of the sample budget added 2.0–3.7 dB of peak signal-to-noise ratio and took 1.3–1.8 times as long. Even the Draft preset (16 samples) exceeded 0.993, which makes it adequate for previews.

The references took 98.6 s and 78.5 s, only two to three times the Standard time despite a 16-fold larger budget. This is consistent with most pixels converging, and stopping, well before 512 samples.

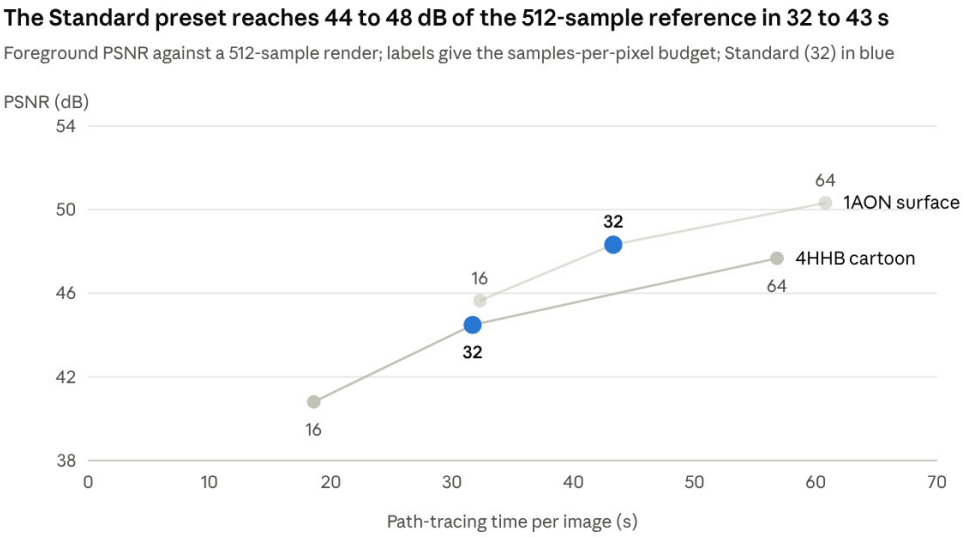
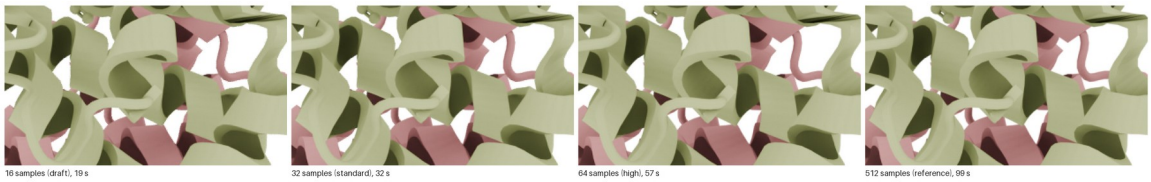


Figure 5 · experiment E3, paper/benchmark/results.csv and summary.json · 4 vCPUs, 1280 × 720

Figure 5. Image fidelity against path-tracing time for the three quality presets. PSNR is computed on luminance over the molecule's pixels (background excluded) against a 512-sample render of the same scene. Whole-image values, dominated by white background, are higher (SSIM 0.997–0.999).



Supplementary Figure S1. The same region of the haemoglobin cartoon at 16, 32, 64 and 512 samples per pixel, enlarged twofold with nearest-neighbour scaling. After denoising the differences are hard to see by eye, consistent with the high similarity scores.

3.5 Threads and image size

Path tracing scaled almost perfectly with CPU threads (experiment E4). The haemoglobin cartoon took 123.7 s on one thread, 61.4 s on two and 30.1 s on four, speed-ups of 2.01 and 4.11, with unchanged peak memory (2.0 GB). Matching the thread count to the container's CPU quota, as the application does, therefore pays off directly.

Render time grew linearly with the number of pixels (experiment E7): 25.3 s at 800 × 800, 31.2 s at 1280 × 720, 44.6 s at 1080 × 1080 and 69.4 s at 1920 × 1080. A straight line fits these points with a slope of 31.3 s per megapixel and an intercept of 5.0 s (R^2 = 0.985). Peak memory rose from 1.8 to 2.9 GB over the same range.

3.6 Renders are bit-for-bit repeatable

Seven renders with identical options, made in separate processes during experiments E2, E4 (with 1, 2 and 4 threads), E6 (two repeats) and E7, produced byte-identical PNG files with the same SHA-256 digest (beginning 6cf024ab). The output therefore depends only on the options and the software versions, not on the run or on the number of threads. This is what justifies naming each image by the hash of its options: a cached image is exactly the image that would be rendered again.

4. Discussion

4.1 Path-traced figures are affordable for almost the whole PDB

The benchmark answers a practical question: what does a path-traced figure cost? Of the 262,874 current PDB entries with a recorded atom count, 97.9% have at most 58,870 deposited atoms, the size of the GroEL–GroES complex, and the median entry has 4,041 atoms, close to haemoglobin (RCSB Data API, 5 November 2026; script and counts in the repository). In that range every image in the benchmark finished within one minute on four CPU threads and used at most 3.6 GB of memory (Section 3.2). Path tracing, once reserved for figures that justified hours of rendering, is therefore cheap enough for a free service on modest cloud hardware.

Two caveats apply. Biological assemblies can be far larger than the deposited coordinates, although instancing keeps their memory low (Section 3.3). And the public instance runs on different hardware (two physical cores, four hardware threads), so its absolute times will differ from those measured here.

4.2 Where the tool fits

Render the Molecule does not replace interactive viewers; it adds the last step that most of them leave to the user (Table 3). Browser viewers such as Mol* and desktop programs such as ChimeraX are where structures are explored, selected and understood, and they export images from real-time rasterisation. Blender-based tools give full control over a physically based renderer but require the Blender application and its learning curve. Render the Molecule sits between these: it offers a fixed menu of scenes rendered with the same path tracer, with no installation and no 3D-graphics expertise.

Table 3. Render the Molecule beside representative tools for making molecular images. Descriptions follow the cited publications.

Tool	Where it runs	How the image is made	Interaction	Typical use
Mol* (Sehnal et al., 2021; Rose et al., 2026)	Browser (WebGL)	Real-time rasterisation with screen-space global illumination	Interactive	Exploring and analysing entries in structure databases
NGL, 3Dmol.js, iCn3D (Rose et al., 2018; Rego & Koes, 2015; Wang et	Browser (WebGL)	Real-time rasterisation	Interactive	Embedding and sharing views

Tool	Where it runs	How the image is made	Interaction	Typical use
al., 2020)				
UCSF ChimeraX (Goddard et al., 2018; Pettersen et al., 2021)	Desktop installation	Real-time rendering with interactive ambient occlusion	Interactive, scripted	Analysis and figures
EzMol (Reynolds et al., 2018)	Web server	Server-side image production after a short wizard	Guided steps	Quick images without expertise
The Protein Imager (Tomasello et al., 2020)	Browser and server	Online viewer linked to automated server-side high-quality rendering	Interactive viewer	Publication illustrations
Prot2Prot (Durrant, 2022)	Browser	Neural image-to-image translation imitating photorealistic styles	Interactive	Fast photorealistic-looking images
BlendMol, Pyrite, Molecular Nodes (Durrant, 2019; Rajendiran & Durrant, 2018; Johnston & Molecular Nodes contributors, 2026)	Desktop Blender	Blender path tracing of any scene the user builds	Full 3D suite	Bespoke figures and animations
Render the Molecule (this work)	Web server, command line, Colab	Cycles path tracing with denoising, preset scenes	Form with presets	Publication images in one step, with embedded provenance

4.3 Provenance and reproducibility

Two properties distinguish the output from a screenshot. First, the image carries its own provenance: the software versions, the source entry and complete references are stored in the PNG file and travel with it when it is copied or downloaded, so a figure can be traced to its origin after it has left the lab. Services that re-encode images can strip these text chunks, which is why the visible credit line is on by default. This puts software citation principles into practice at the level of the individual image (Smith et al., 2016; Katz et al., 2021) and makes the image itself a findable, attributable research object in the sense of the FAIR principles (Wilkinson et al., 2016; Barker et al., 2022).

Second, the image is reproducible. Renders were byte-identical across processes and thread counts (Section 3.6), because the path tracer's random seed is fixed and the CPU denoiser returned identical output in every repeat. Whether this holds across processor models and GPU back ends was not tested.

Together with exact dependency pins and a containerised runtime (Boettiger, 2015; Grüning et al., 2018), this means the triplet of PDB entry, option set and software version determines the pixels. It also makes content-addressed caching safe, which is what keeps a free public service affordable: a popular image is computed once.

4.4 Limitations

Render the Molecule trades control for simplicity. The camera offers seven preset viewpoints and no free rotation, zoom or framing, so a user cannot yet centre a binding site or crop to a domain. Selections are fixed: the tool cannot highlight chosen residues, label atoms, or colour by B-factor, hydrophobicity or electrostatic potential. Users who need these should prepare the scene in an interactive viewer or in Blender itself.

Input is limited to entries in the PDB archive. Computed structure models from AlphaFold DB and ModelArchive (Varadi et al., 2024; Bittrich et al., 2023), integrative structures and unpublished coordinates cannot yet be rendered through the web form, although the command-line tool accepts a local BinaryCIF file.

The palette needs care. The rainbow scheme helps a reader follow a chain from N- to C-terminus, but rainbow colour maps are not perceptually uniform and are hard to read for people with colour-vision deficiency (Crameri et al., 2020). The per-chain colours have not yet been tested for colour-vision deficiency.

The atom limits are miscalibrated in both directions. They are too strict for assemblies, because they count every copy although memory does not grow with the number of copies (Section 3.3). They are too lenient for large asymmetric units: a ribosome surface within the surface limit needed 10.2 GB, more than the public deployment provides, and a 2.44-million-atom cartoon within the cartoon limit exhausted 15 GB. They are also checked late, after a parse that took four minutes for the largest entry. Limits derived from measured memory per representation, checked against the deposited atom count from the RCSB Data API before download, would refuse hopeless requests in seconds and admit large assemblies.

The public service is deliberately small. It renders one image at a time on two physical CPU cores, keeps its queue in memory (lost on a container restart) and starts from zero after a quiet period, so the first visitor waits for the container to start. The mandatory name and email address let the operator see who uses the service, but they are personal data, add friction, and may count as a login requirement for some journals' web-server issues. Finally, the tool depends on Molecular Nodes, whose interface changes between releases; pinning every dependency keeps renders reproducible but means upgrades must be deliberate and re-tested.

4.5 Future work

Four extensions follow from these limits. First, accepting a scene description, for example a MolViewSpec state exported from Mol* (Midlik et al., 2025), would let users choose the view, selections and colours interactively and then obtain the path-traced image. Second, accepting AlphaFold DB and ModelArchive identifiers and uploaded files would cover computed and integrative models, including predicted complexes (Abramson et al., 2024). Third, rendering on a GPU, which Cycles supports, would

shorten render times where one is available. Fourth, colour-vision-safe palettes and per-residue colouring would make the images more accessible and more informative. A user study comparing comprehension of path-traced and rasterised figures would test the premise behind the tool directly.

5. Conclusion

Render the Molecule turns a PDB identifier into a path-traced, citable image without any installation, by running Blender and Molecular Nodes as a headless service. For the 97.9% of PDB entries with at most about 59,000 atoms, an image costs under a minute on four CPU threads and under 4 GB of memory, and the same request always returns the same bytes, so content-addressed caching serves repeats exactly and for free. Embedded metadata keeps the software versions, the source entry and the references attached to the file wherever it travels. The tool complements interactive viewers: users explore and choose in Mol*, ChimeraX or PyMOL, and render the final figure here.

Availability and requirements

Item	Detail
Project name	Render the Molecule
Web application	vishalbhoir18--render-the-molecule.us-east.modal.run
Landing page and Colab notebook	vishalbhoir18.github.io/renderthemolecule
Source code	github.com/vishalbhoir18/renderthemolecule
Archived version	v0.1.0, doi:10.5281/zenodo.23960655
Operating system	Any modern browser; self-hosting on Linux x86-64 (Docker)
Programming language	Python 3.13
Main dependencies	Blender 5.2.2 (bpy), Molecular Nodes 520.2.0, FastAPI, Pillow
Licence	GNU GPL v3 or later
Restrictions for use	None; users give a name and email address before rendering

Data availability

The benchmark and analysis scripts, the per-render results table (results.csv), the image-quality summary (summary.json), the atom counts of all current PDB entries (pdb_atom_counts.csv) and the figure files are in the paper folder of the repository. The 60 benchmark images (25 MB) are regenerated by the benchmark script and should be identical with the pinned software versions (Section 3.6); they will be deposited with the next Zenodo release. Structure data are from the RCSB PDB.

Declarations

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Author contributions (CRediT). Vishal Bhoir: conceptualization, software, validation, investigation, writing (original draft, review and editing). To be confirmed by the author.

Use of AI tools. Drafting of this manuscript, the benchmark scripts and the literature search were assisted by an AI assistant (Claude, Anthropic). The author reviewed and takes responsibility for all content. Most journals require a statement of this kind; adapt it to the target journal's policy.

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Every journal reference was checked against Crossref by its DOI on 5 November 2026. Entries follow APA 7 and should be reformatted to the target journal's style.

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